

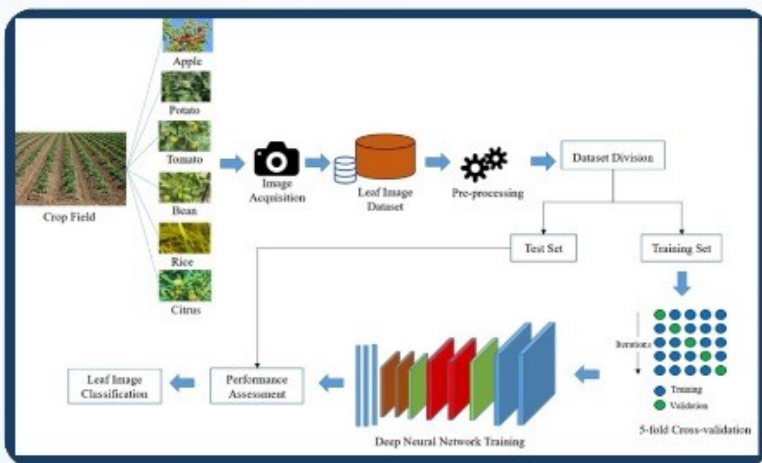


CENTRE FOR ADVANCED STUDIES
LUCKNOW, UTTAR PRADESH, INDIA

INSPIRING STUDENT INNOVATION

Dense Convolutional Neural Networks based Multiclass Plant Disease Detection and Classification

Students of Centre for Advanced Studies have jointly developed a deep-learning-based approach for the plant disease detection & classification from leaf images captured in various resolutions. Dense convolutional neural network architecture is trained on a large plant leaves image dataset (both laboratory and on-field conditions) of six crops in 27 different categories from multiple countries. Images have several inter-class and intra-class variations with complex and challenging conditions that have been addressed in this dense neural network. Five-fold cross-validation & testing on unseen data is done for exhaustive evaluation of the trained model in various parameters. The experimental findings demonstrate that an average test accuracy of 99.199% is obtained on unseen images with complex background conditions. The processing time to process a single plant leaf image is 0.016s with significant accuracy which signifies its real-time performance.



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As plant diseases are one of the major concern in the agricultural domain, this research work will make good contribution in the field of plant disease detection and their classification using digital images to take the necessary action by farmers and plant pathologists.

RESEARCH TEAM



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RESEARCH PUBLICATION FROM THIS WORK

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